

# FOX\_Checklist

**What this is:** every FoxByte feature and its sub-features — built or not, and how it was tested — so it's clear where we are and what's next. How each feature works is in `FOX_Feature_Implemented.pdf` (same ids).

**Last updated:** 22 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection ( `chore/foxbyte` )

**Source:** `docs/FOX_Checklist.html` → `make feature-doc` → this PDF

✔ Done & tested — automated tests (unit, VM integration or CI)

🟡 Partial — gaps, or not verified end to end

⚠ Sub-features: ✔ built 🟡 partly ! a known limit → planned, not built □ not built yet ✖ broken

✔ Done — tested by hand only

✖ Known issue

□ Not started

The badge rule

(audit v2 G31, which found 47 rows badged done while carrying an open item): a row is *done* or *done (manual)* only when nothing in it is 🟡, □ or ✖, and `scripts/test_feature_checklist.sh` fails the build otherwise. A ! is a limit of what the feature does and a → is a planned extension — neither makes the row's own claim untrue. How thoroughly something was tested belongs in "Tested by", not in the sub-features. · "VM §n" = section of `make integration` ;"v2 §n" = `make integration-v2` ;"update" = `make integration-update`

## Where we are

| Area                                    | Features | Done & tested | Done (manual) | Partial | Known issue | Not started | Progress                    |
|-----------------------------------------|----------|---------------|---------------|---------|-------------|-------------|-----------------------------|
| A · Install & setup                     | 9        | 4             | 3             | 2       | ·           | ·           | <div><div></div></div> 78%  |
| B · CLI & services                      | 3        | 3             | ·             | ·       | ·           | ·           | <div><div></div></div> 100% |
| C · Branching & storage                 | 8        | 7             | ·             | 1       | ·           | ·           | <div><div></div></div> 88%  |
| D · Durability                          | 6        | 5             | ·             | 1       | ·           | ·           | <div><div></div></div> 83%  |
| E · Security foundations                | 3        | 3             | ·             | ·       | ·           | ·           | <div><div></div></div> 100% |
| F · SQL gateway                         | 7        | 6             | 1             | ·       | ·           | ·           | <div><div></div></div> 100% |
| G · Accounts & auth                     | 6        | 5             | ·             | 1       | ·           | ·           | <div><div></div></div> 83%  |
| H · Blackbox (record of schema changes) | 18       | 16            | ·             | 2       | ·           | ·           | <div><div></div></div> 89%  |
| I · Import & migration                  | 13       | 5             | 8             | ·       | ·           | ·           | <div><div></div></div> 100% |
| J · ETL pipelines (experimental)        | 4        | 2             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| K · APIs, SDKs & agents                 | 6        | 4             | 2             | ·       | ·           | ·           | <div><div></div></div> 100% |
| L · Web console                         | 11       | 1             | 10            | ·       | ·           | ·           | <div><div></div></div> 100% |
| M · Build, CI & releases                | 7        | 6             | 1             | ·       | ·           | ·           | <div><div></div></div> 100% |
| Total (features built or started)       | 101      | 67            | 27            | 7       | ·           | ·           | <div><div></div></div> 93%  |
| N · Not started (roadmap)               | 8        | 1             | ·             | ·       | ·           | 7           | <div><div></div></div> 13%  |

Storage engine — not implemented, by design

FoxByte does **not** have its own storage engine. Data is stored by **PostgreSQL's own engine** (heap tables, B-tree indexes, write-ahead log) running on local disk inside the stock `postgres:18-bookworm` image plus `wal-g` ( `postgres:16-bookworm` for an install created before 21 Sep 2026). This was a locked architecture decision: a Neon-style engine that separates compute from storage over the network adds latency to every transaction, and transaction speed must never be compromised.

What **is** built is a storage *layer* around Postgres: copy-on-write branch drivers — ZFS on macOS/Linux, btrfs on Windows — behind one interface in `internal/branch/storage.go` (C1), quotas (C6), capacity reporting (C7), and durability to object storage with `wal-g` + MinIO: WAL archiving, base backups, point-in-time restore (D1–D3). Networked storage disaggregation is listed in the README roadmap but not started (see "Not started").

## Suggested next steps

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1. **The claims are checkable now:** a restore is proved weekly and by `fox backup verify` (D2), a tag cannot ship past a red or stale nightly (M2), every command the engine runs has a deadline so nothing hangs for ever (G30's useful half), containers restart on crash (B2), and CI runs the race detector. Before this, the most consequential thing the product claims — that the backups restore — had never been tested end to end on a real install.
2. **This checklist's own badge rule is enforced** (audit v2 G31): no row can be badged done while carrying an open sub-item, and `scripts/test_feature_checklist.sh` fails the build if one is. The 45 sub-items that broke the rule were each put somewhere honest — built, a limit (!), planned (→), or a note in "Tested by" — and nine of them turned out to be things we had already shipped and never ticked off, including automated UI tests, per-user access control, SDK contract tests and the first signed release.
3. Run on real machines what the throwaway VM cannot: `fox update` across a real release on macOS and Windows (A9); on Windows, the `zstd` distro image's unpack (M5), the `btrfs` holding area of `fox pg upgrade` (D6), and an export's copy to the host (D5).
4. Automated tests for the imports (I1–I3, I7) — the last area with none.
5. Gate release tags on the nightly suites, so a tag cannot ship past a red suite (M2).
6. Decide how this checklist shows partial work (audit v2 G31): 9 features are badged done while carrying an open sub-item ( C7, D5, D6, F5, G3, H16, H17, I13, M7 ) — down from 34. Each of those sub-items is a disclosed limit rather than unfinished work, so the remaining question is whether a limit belongs in the sub-item list at all.
7. **Promotion is built** (H18, audit G29): a branch's recorded statements, reviewed, then applied to another branch in one transaction — attributed to whoever wrote them, with the approver beside them. Schema only. An agent can propose and cannot approve its own request.
8. **From the audit, deliberately left:** HA beyond the demo — automatic failover and fencing across hosts (D4, G21); metrics and alerts, and supervision of the background servers themselves (B2, G27 — the containers restart); splitting the branch package and running its SQL through `pgx` (G30 — the deadlines are in); quotas on `btrfs` and per-branch CPU/memory (C6, G33); `pgvector` in the image (M6, G34); `/api/v1` (K1, G35).

| ID                                                                                     | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Tested by                                                                            |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  A1   | <b>One-line installer (macOS / Linux)</b> <ul style="list-style-type: none"> <li>✓ Detects OS / CPU, downloads the release binary</li> <li>✓ Installs to the prefix (sudo only if needed)</li> <li>✓ Stages the Linux engine for the VM on macOS</li> <li>✓ <code>FOX_VERSION</code>, <code>FOX_REPO</code>, <code>FOX_DIST</code>, <code>FOX_PREFIX</code></li> <li>✓ Verifies every download against <code>SHA256SUMS</code>; refuses what it can't check</li> <li>✓ Release signature checked (OpenSSL 3; warns with LibreSSL)</li> </ul>                                                                                                                                                      | Unit ( <code>test_install_sh.sh</code> , CI); manual                                 |
|  A2   | <b>Windows installer (install.ps1)</b> <ul style="list-style-type: none"> <li>✓ Installs WSL, elevates, resumes after reboot</li> <li>✓ Downloads fox.exe, engine, image build context</li> <li>✓ Ubuntu rootfs verified against Ubuntu's checksums</li> <li>✓ Adds to PATH and runs <code>fox setup</code></li> <li>✓ Prefers the prebuilt distro image, falls back to the rootfs</li> <li>✓ Verifies FoxByte's own downloads</li> <li>✓ Run on a real Windows machine (distro image downloaded and verified)</li> </ul>                                                                                                                                                                         | Pester (Windows CI); Windows run 16 Sep                                              |
|  A3   | <b>fox setup on macOS (Lima)</b> <ul style="list-style-type: none"> <li>✓ Creates / starts the Lima VM</li> <li>✓ Installs Docker and ZFS</li> <li>✓ Reinstalls the engine on every run</li> <li>✓ Starts the stack</li> <li>✓ Repairs an interrupted provisioning on the next run (dpkg interrupted, half-installed packages)</li> <li>✓ Image builds from the copy inside fox when it cannot be pulled (C8)</li> </ul>                                                                                                                                                                                                                                                                          | Unit; run against a deliberately broken VM, 21 Sep                                   |
|  A4   | <b>fox setup on Windows (WSL2 + btrfs)</b> <ul style="list-style-type: none"> <li>✓ Imports a dedicated distro with systemd</li> <li>✓ Docker, btrfs and a storage mount unit</li> <li>✓ Quiet steps, detail in install.log</li> <li>✓ Waits for systemd on a cold distro</li> <li>✓ Preloaded images tagged as the engine runs them</li> <li>✓ Summary with API key and <code>sslmode</code></li> <li>✓ Fresh install from v0.8.6 on Windows: preload used, no apt, key in the summary</li> </ul>                                                                                                                                                                                                | Unit ( <code>host_wsl_test.go</code> ); Windows run 16 Sep                           |
|  A5 | <b>Linux native</b> <ul style="list-style-type: none"> <li>✓ Engine runs in-process, no VM</li> <li>✓ <code>fox setup</code> explains nothing is needed</li> <li>✓ Services listen on loopback by default ( <code>FOX_LISTEN</code> to expose)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         | VM integration (all suites run on Linux)                                             |
|  A6 | <b>Command forwarding &amp; host file streaming</b> <ul style="list-style-type: none"> <li>✓ macOS → Lima VM, starts the VM if stopped</li> <li>✓ Windows → WSL distro</li> <li>✓ <code>fox import</code> streams a host file over stdin → <a href="#">Stream pipeline run spec files too</a></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                          | Unit ( <code>TestImportLocalFile</code> , <code>TestWSLArgs</code> )                 |
|  A7 | <b>Always-latest engine binary</b> <ul style="list-style-type: none"> <li>✓ Downloads the newest engine into the cache (ELF check)</li> <li>✓ <code>FOX_NO_REFRESH</code>, <code>FOX_VERSION</code></li> <li>✓ Verified against <code>SHA256SUMS</code>; keeps the staged engine if not</li> <li>✓ Seen verifying on a real Windows install</li> </ul>                                                                                                                                                                                                                                                                                                                                            | Unit ( <code>host_test.go</code> , <code>update_test.go</code> ); Windows run 16 Sep |
|  A8 | <b>New-release notice on fox start</b> <ul style="list-style-type: none"> <li>✓ Background check, 1.5 s limit, printed last</li> <li>✓ Only a promoted, newer, complete release</li> <li>✓ One request per start; a found release remembered 6 h, never "up to date"</li> <li>✓ Off switch; never on dev builds; silent offline</li> </ul>                                                                                                                                                                                                                                                                                                                                                        | Unit; update §1, §1b, §2, §3, §8; real Mac check                                     |
|  A9 | <b>fox update</b> <ul style="list-style-type: none"> <li>✓ Downloads and verifies everything before changing anything</li> <li>✓ Stops servers, installs engine, restarts old one on failure</li> <li>✓ <code>up</code>, Blackbox upgrade, <code>start</code> on the new engine</li> <li>✓ Replaces host <code>fox</code>, keeps previous binaries</li> <li>✓ <code>--check</code>, <code>--yes</code>, <code>--version</code></li> <li>✓ Linux path verified end to end</li> <li>✓ macOS path run against real releases (v0.8.3 → v0.8.4 → v0.8.5)</li> <li>❑ Windows path run against a real release (deferred)</li> <li>✓ Release signatures (Ed25519 over <code>SHA256SUMS</code>)</li> </ul> | Unit; update §4–§7; a real Mac run                                                   |

| ID | Feature · sub-features | Tested by |
|----|------------------------|-----------|
|----|------------------------|-----------|

|   |                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                             |
|---|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ✓ | <b>B1 The fox command set</b>                       | <ul style="list-style-type: none"> <li>✓ Stack: setup, start, stop, up, down, status, check, logs, connect, psql, demo, update</li> <li>✓ Branching, backup / restore, HA</li> <li>✓ Blackbox, policy, impact</li> <li>✓ Import, pipelines</li> <li>✓ backup export / restore, pg status / upgrade</li> </ul>                                                                                                                                                                                                                    | <ul style="list-style-type: none"> <li>✓ Users, API keys, admin</li> <li>✓ <code>fox vm status / fox vm shell</code> (Lima, WSL; Linux says there is no VM)</li> <li>✓ <code>branch create --from</code></li> <li>✓ <code>fox uninstall</code> removes it all, retired installs included</li> </ul>                                                                                                                                                                                                                                                                                                         | Unit ( <code>main_test.go</code> , <code>update_test.go</code> ); VM suites |
| ✓ | <b>B2 Background services (daemon)</b>              | <ul style="list-style-type: none"> <li>✓ Control plane :8080, gateway :6432, agent API :8088</li> <li>✓ Detached, pidfiles and logs in <code>~/ .fox</code></li> <li>✓ Stop (TERM then KILL), status, logs</li> </ul>                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>✓ Containers restart on crash and after a reboot ( <code>--restart=unless--stopped</code> ); a container stopped on purpose stays stopped</li> <li>! The three background servers themselves are not supervised ( <code>fox check</code> reports them)</li> </ul>                                                                                                                                                                                                                                                                                                    | Unit ( <code>TestStopTerminatesProcess</code> ); VM §1                      |
| ✓ | <b>B3 First run (sample data, check , connect )</b> | <ul style="list-style-type: none"> <li>✓ Sample data seeded into <code>main</code> on a first start: <code>users</code> , <code>projects</code> , <code>events</code> , five rows each, two foreign keys</li> <li>✓ <code>fox demo seed drop sql</code> on any branch; <code>FOX_NO_DEMO=1</code> starts empty</li> <li>✓ The console opens itself, with the one-time setup token already in the link</li> <li>✓ Declines to open a browser in the VM, over SSH, with no display, or with <code>FOX_NO_BROWSER</code></li> </ul> | <ul style="list-style-type: none"> <li>✓ <code>fox check</code> : version, console, storage, main, three servers, three ports, TLS, Blackbox, backup age, sample data — each failure with the command that fixes it</li> <li>✓ <code>fox connect [branch]</code> prints the application's connection string, then opens psql; <code>fox psql</code> takes a branch</li> <li>✓ <code>fox branch create</code> ends with the connection string and how long it took</li> <li>✓ Plain-English errors in the CLI and the console ( <code>internal/errhint</code> , <code>web/src/errhint.ts</code> )</li> </ul> | Unit; v1 §11d                                                               |

## C Branching & storage

7 of 8 done

| ID   | Feature · sub-features                                                                                                                                                                                                                            | Tested by                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| ✓ C1 | <b>Storage drivers (ZFS / btrfs)</b> <ul style="list-style-type: none"> <li>✓ ZFS snapshot + clone per branch (macOS VM, Linux)</li> <li>✓ btrfs subvolume snapshots (Windows)</li> <li>✓ Pool / filesystem created on a loopback file</li> </ul> | Unit; VM §2, §2b (ZFS); btrfs manual on Windows                             |
| ✓ C2 | <b>Branch lifecycle</b> <ul style="list-style-type: none"> <li>✓ Create, list, delete</li> <li>✓ Reset to parent</li> <li>✓ Suspend / resume</li> <li>✓ Recreate a container from its surviving data</li> </ul>                                   | Unit; VM §2, §2b, §4                                                        |
| ✓ C3 | <b>Protection of main</b> <ul style="list-style-type: none"> <li>✓ Delete and reset refused</li> </ul>                                                                                                                                            | Unit; VM §4, §6                                                             |
| ✓ C4 | <b>Private networking (no host ports)</b> <ul style="list-style-type: none"> <li>✓ Branch databases reachable only through the gateway</li> <li>✓ Published ports (object store, restore, debug) on loopback only; standby unpublished</li> </ul> | Unit ( <code>TestParsePublishedPort</code> , <code>TestDSN</code> ); VM §1c |
| ✓ C5 | <b>Scale-to-zero &amp; wake on connect</b> <ul style="list-style-type: none"> <li>✓ Idle branches suspended automatically</li> <li>✓ Next connection wakes the branch (~0.3–0.7 s)</li> </ul>                                                     | VM §7                                                                       |
| 🔗 C6 | <b>Storage quotas</b> <ul style="list-style-type: none"> <li>✓ Space reserved for main (ZFS)</li> <li>🔗 Per-branch cap (opt-in)</li> </ul>                                                                                                        | Manual                                                                      |
| ✓ C7 | <b>Capacity &amp; status</b> <ul style="list-style-type: none"> <li>✓ Services, main health, backups, branches</li> <li>✓ Storage used / available</li> </ul>                                                                                     | VM §1                                                                       |

|   |                                                                                                                                                                                                                                         |               |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| ✓ | <b>C8 Provisioning &amp; image pull</b>                                                                                                                                                                                                 | Unit; VM §11b |
| ✓ | <ul style="list-style-type: none"> <li>Docker, storage pool and image prepared on up</li> <li>Pulls from GHCR, builds locally as fallback</li> <li>The build context is built into fox, so the fallback works on any install</li> </ul> |               |

D Durability

5 of 6 done

| ID                                         | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Tested by                                                                        |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| ✓ D1                                       | <b>WAL archiving to object storage</b> <ul style="list-style-type: none"> <li>wal-g ships main's WAL to MinIO, off the commit path</li> <li>MinIO images pinned on quay.io</li> <li>Branches are disposable by design (export one that matters)</li> <li>Remote S3 target ( <code>fox backup target set</code> ), keys in 0600 env files</li> <li>Object Lock reported; tested on MinIO</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          | VM §3; VM §11c; not yet run against AWS S3                                       |
| ✓ D2                                       | <b>Base backups</b> <ul style="list-style-type: none"> <li>Create and list</li> <li>Follow the primary after a failover; listing works with main stopped</li> <li>Read-only listing: <code>GET /api/backups</code> and a Dashboard panel</li> <li>Schedule and retention ( <code>FOX_BACKUP_INTERVAL</code> , <code>FOX_BACKUP_RETAIN</code> )</li> <li>A first backup at install</li> <li>Age shown and warned in <code>fox status</code> and the status API</li> <li>Proven, not assumed: <code>fox backup verify</code> writes a probe row, waits for its WAL segment to be archived, restores the newest base backup into a container of its own and reads the row back</li> <li>On a schedule (weekly; <code>FOX_RESTORE_CHECK_INTERVAL</code> ), with the result in <code>fox check</code></li> </ul> | VM §3, §3b, §6; VM §11c                                                          |
| ✓ D3                                       | <b>Point-in-time restore</b> <ul style="list-style-type: none"> <li>Restore to a timestamp or latest on :5433</li> <li>Per-install credentials</li> <li>Starts from the newest base backup that precedes the target</li> <li>A point no backup precedes is refused, naming the oldest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Unit; VM §3, §3b, §6; v2 §1a                                                     |
| 🔥 D4                                       | <b>HA standby &amp; failover</b> <ul style="list-style-type: none"> <li>Streaming hot standby</li> <li>Manual failover, <code>main</code> rerouted</li> <li>Nothing deletes or stops the promoted standby</li> <li><code>fox up</code> brings it back, not the old main</li> <li><code>fox ha fallback</code>, keeping every write</li> <li>The promoted standby archives WAL, so backups and PITR cover the failover</li> <li>Automatic failover, fencing, multi-host</li> </ul>                                                                                                                                                                                                                                                                                                                           | Unit; VM §6, §7                                                                  |
| ✓ D5                                       | <b>Portable export &amp; restore</b> <ul style="list-style-type: none"> <li>Every non-agent branch: <code>pg_dump</code>, its roles, the Blackbox head; SHA256SUMS over all</li> <li>Copied out of the VM and verified on the host; a copy that fails is deleted</li> <li>Damaged, incomplete or path-escaping archives refused</li> <li>Restore into a new branch, same or newer major; Blackbox checked exact</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                  | Unit; VM pg-upgrade §1; the host copy on macOS/Windows is unit-tested only       |
| ✓ D6                                       | <b>PostgreSQL major upgrade</b> <ul style="list-style-type: none"> <li><code>fox pg status</code>: the major of the install and of each branch</li> <li>Plan with every change crossed, found-in-your-data first; refusals say what to do</li> <li>Asks, exports, then reloads every carried branch on the new major</li> <li>Data, roles, API keys and every Blackbox entry come across</li> <li>Old databases held outside the branch namespace; rollback and finalize</li> <li>Each upgraded cluster archives WAL under its own prefix</li> </ul>                                                                                                                                                                                                                                                        | Unit; VM pg-upgrade (72 / 72); the btrfs (Windows) holding area has not been run |
| → Carry agent branches; upgrade with HA on |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                  |

E Security foundations

3 of 3 done

| ID   | Feature · sub-features                                                                                                                                                                                                                                                             | Tested by                       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| ✓ E1 | <b>Per-install generated secrets</b> <ul style="list-style-type: none"> <li>Generated Postgres and MinIO passwords</li> <li>Used by restore too</li> <li>Each login role has its own derived password</li> <li>Never on a command line (docker <code>--env-file</code>)</li> </ul> | Unit (indirect); v2 §1a; VM §1d |

|   |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                     |
|---|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| ✓ | <b>E2 TLS everywhere</b>                          | <ul style="list-style-type: none"> <li>✓ Self-signed TLS on all three services, zero setup</li> <li>✓ TLS required from other machines (loopback clients may skip it)</li> <li>✓ No plain-HTTP fallback except on loopback</li> </ul>                                                                                                                                                                                   | Unit<br>( <code>TestReadStartupUpgradesTLS</code> ) |
| ✓ | <b>E3 Keys are made by a person, not by start</b> | <ul style="list-style-type: none"> <li>✓ Start creates no account and no key, and caches none on disk</li> <li>✓ The banner shows the connection string and how to make a key</li> <li>✓ The first account created gets the admin grant on main</li> <li>✓ First account over HTTP needs the setup token ( <code>fox_setup-token</code> )</li> <li>✓ Exactly one first account, however many sign up at once</li> </ul> | Integration §1                                      |

## F SQL gateway

7 of 7 done

| ID | Feature · sub-features                                                                                                                                                                          | Tested by                                                                                                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓  | <b>F1 Routing by database name</b> <ul style="list-style-type: none"> <li>✓ One endpoint :6432, database name = branch</li> </ul>                                                               | Unit; VM §1, §6<br>→ <a href="#">Connection pooling, rate limiting</a>                                                                                                       |
| ✓  | <b>F2 API-key authentication</b> <ul style="list-style-type: none"> <li>✓ <code>key_</code> key as the Postgres password</li> <li>✓ Bad key rejected with a Postgres error</li> </ul>           | VM §1<br><ul style="list-style-type: none"> <li>✓ Startup deadline and connection cap</li> <li>✓ An account key opens main, its own branches, or any for an admin</li> </ul> |
| ✓  | <b>F3 Backend login (SCRAM)</b> <ul style="list-style-type: none"> <li>✓ Cleartext, md5, SCRAM-SHA-256</li> </ul>                                                                               | VM §1, §6, §8 (exercised); no unit test for SCRAM yet<br><ul style="list-style-type: none"> <li>✓ Clients never see the database password</li> </ul>                         |
| ✓  | <b>F4 Per-user Postgres login roles</b> <ul style="list-style-type: none"> <li>✓ Role per user, actor can't be forged</li> <li>✓ Permissions of non-superuser <code>db_client</code></li> </ul> | Manual<br><ul style="list-style-type: none"> <li>✓ Own password per role</li> </ul>                                                                                          |
| ✓  | <b>F5 Attribution session settings</b> <ul style="list-style-type: none"> <li>✓ Actor kind, branch, session id per connection</li> </ul>                                                        | VM §8<br>! <a href="#">Kind / branch / session client-settable</a>                                                                                                           |
| ✓  | <b>F6 Ctrl-C (CancelRequest) forwarding</b> <ul style="list-style-type: none"> <li>✓ Cancel a running query through the gateway</li> </ul>                                                      | Unit                                                                                                                                                                         |
| ✓  | <b>F7 Errors &amp; insecure build tag</b> <ul style="list-style-type: none"> <li>✓ Real Postgres error responses</li> </ul>                                                                     | Unit ( <code>TestSendError</code> )<br><ul style="list-style-type: none"> <li>✓ No-auth bypass impossible in release builds</li> </ul>                                       |

## G Accounts & auth

5 of 6 done

| ID | Feature · sub-features                                                                                                                                                                                                                               | Tested by                                                                                                                                                             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓  | <b>G1 Auth store (SQLite)</b> <ul style="list-style-type: none"> <li>✓ Users, keys, sessions, OAuth ids, pipelines</li> <li>✓ Safe under concurrent use (WAL, busy timeout), including a new store created by three services at once</li> </ul>      | Unit ( <code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code> ); VM §1<br>→ <a href="#">Roles / permissions, multi-tenant projects</a>           |
| ✓  | <b>G2 Email &amp; password users</b> <ul style="list-style-type: none"> <li>✓ Create account, log in</li> <li>✓ Password change (signs out other sessions); admin reset at the CLI</li> <li>→ <a href="#">Self-service reset by email</a></li> </ul> | Unit; VM §1; VM §1d<br><ul style="list-style-type: none"> <li>✓ List and delete users (admins)</li> <li>✓ Emails validated; at most 63 bytes (a role name)</li> </ul> |
| ✓  | <b>G3 API keys</b> <ul style="list-style-type: none"> <li>✓ Create (shown once), list, revoke</li> <li>✓ Bearer / X-API-Key and gateway password</li> </ul>                                                                                          | Unit; VM §1<br><ul style="list-style-type: none"> <li>✓ Revoking an id that is not there is refused (404), not reported as revoked</li> </ul>                         |
| ✓  | <b>G4 Web sessions &amp; signup control</b> <ul style="list-style-type: none"> <li>✓ httpOnly session cookie</li> <li>✓ Sign-up closed by default ( <code>FOX_SIGNUP=open</code> )</li> </ul>                                                        | Unit ( <code>TestSessions</code> )<br><ul style="list-style-type: none"> <li>✓ Failed sign-ins throttled</li> </ul>                                                   |

|                                                                                   |                                   |                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                   |              |
|-----------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|  | <b>G5 OAuth (GitHub / Google)</b> | <ul style="list-style-type: none"> <li>✓ Sign-in flows built</li> <li>✓ Defaults match what the engine serves (<code>https://localhost:8080</code> ; web origin = public URL)</li> <li>✓ Session cookie <code>Secure</code> / <code>SameSite</code> follow the origin</li> </ul> | <ul style="list-style-type: none"> <li>☐ Tested with a real OAuth app</li> <li>✓ Only a provider-verified email joins or creates an account; closed sign-up respected</li> <li>✓ State cookie one-use, <code>Secure</code> , <code>SameSite=Lax</code></li> </ul> | Unit; VM §1b |
|-----------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|

|   |                        |                                                                                                                                                                                     |                                                                                                                                                                             |              |
|---|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| ✓ | <b>G6 Security log</b> | <ul style="list-style-type: none"> <li>✓ Sign-ins, failures, keys, passwords, accounts, admin grants, refusals</li> <li>✓ Hash-chained, append-only, anchored and signed</li> </ul> | <ul style="list-style-type: none"> <li>✓ <code>fox audit</code> , <code>verify</code> , <code>GET /api/audit</code> → Included in <code>fox backup export</code></li> </ul> | Unit; VM §1d |
|---|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|

## H Blackbox (record of schema changes)

16 of 18 done

| ID                                                                                             | Feature · sub-features                                                                                                                                                                                                                                                                                       | Tested by                             |
|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|  <b>H1</b>    | <b>DDL recording</b> <ul style="list-style-type: none"><li>✓ Who, tool, branch, statement, status, risk</li><li>✓ Inherited by every branch</li><li>✓ A restart records nothing: the install script's own statements are not captured</li><li>! Role / database changes, some dropped object types</li></ul> | VM §8; v2 §9; v2 §10                  |
|  <b>H2</b>    | <b>Hash chain, append-only &amp; verify</b> <ul style="list-style-type: none"><li>✓ Rows can't be edited or deleted normally</li><li>✓ fox ledger verify</li><li>🔴 Superuser rewrite caught only via anchors (H10–H12)</li></ul>                                                                             | VM §8; v2 §0, §6b                     |
|  <b>H3</b>  | <b>Attribution</b> <ul style="list-style-type: none"><li>✓ Humans via their login role</li><li>✓ Agents via their own agent-&lt;id&gt; role</li></ul>                                                                                                                                                        | Unit; VM §8; v2 §1c                   |
|  <b>H4</b>  | <b>Destructive-DDL guardrail</b> <ul style="list-style-type: none"><li>✓ DROP TABLE / DROP SCHEMA refused, attempt recorded</li><li>✓ Override only superusers / db_admin , use recorded</li></ul>                                                                                                           | VM §8; v2 §1b, §6b                    |
|  <b>H5</b>  | <b>Risk flags</b> <ul style="list-style-type: none"><li>✓ Type change, drop column flagged</li><li>✓ Judged on the running statement (no comments or literals)</li></ul>                                                                                                                                     | Unit; v2 §6b                          |
|  <b>H6</b>  | <b>Blackbox access (CLI / REST / MCP / UI / SDK)</b> <ul style="list-style-type: none"><li>✓ CLI, REST with filters, MCP, web page, SDKs</li></ul>                                                                                                                                                           | v2 §0, §5                             |
|  <b>H7</b>  | <b>fox ledger revert --to</b> <ul style="list-style-type: none"><li>✓ Help and output say it's a point-in-time restore of main</li></ul>                                                                                                                                                                     | Unit (help text); VM §3 (the restore) |
|  <b>H8</b>  | <b>Override permission (db_admin) &amp; upgrade</b> <ul style="list-style-type: none"><li>✓ fox admin grant revoke list</li><li>✓ First user granted automatically</li></ul>                                                                                                                                 | v2 §1b, §6                            |
|  <b>H9</b>  | <b>Blackbox 2.0 capture</b> <ul style="list-style-type: none"><li>✓ Transaction id and WAL position per entry</li><li>✓ Task, tool-call hash, override use</li></ul>                                                                                                                                         | Unit; v2 §2                           |
|  <b>H10</b> | <b>Checkpoints &amp; anchors</b> <ul style="list-style-type: none"><li>✓ Merkle root in the database and in read-only anchor files</li><li>✓ On a schedule and on demand</li><li>✓ Published anchor format</li></ul>                                                                                         | Unit; v2 §3, §3b                      |

|   |                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ | <b>H11 fox-verify (independent verifier)</b>                                                                                                                                                                                                                           | Unit; v2 §3                                                                                                                                                                                                                                                                                                                                                        |
| ✓ | <ul style="list-style-type: none"> <li>Checks a live branch or an export</li> <li>Release binaries for every platform</li> </ul>                                                                                                                                       | <ul style="list-style-type: none"> <li>✓ <code>--pubkey</code> checks signatures</li> </ul>                                                                                                                                                                                                                                                                        |
| ✓ | <b>H12 Blackbox integrity &amp; export</b>                                                                                                                                                                                                                             | v2 §3                                                                                                                                                                                                                                                                                                                                                              |
| ✓ | <ul style="list-style-type: none"> <li>INTACT / TAMPERED via CLI, REST, MCP</li> <li>JSON-lines export</li> </ul>                                                                                                                                                      | → Streaming for very large ledgers                                                                                                                                                                                                                                                                                                                                 |
| ✓ | <b>H13 Branch from just before a Blackbox entry</b>                                                                                                                                                                                                                    | Unit; v2 §4; VM ui                                                                                                                                                                                                                                                                                                                                                 |
| ✓ | <ul style="list-style-type: none"> <li>Restore + replay up to the change's transaction</li> <li>By transaction id or timestamp</li> <li>CLI, REST, MCP, the Integrity panel, and every row of the Blackbox page</li> </ul>                                             | <ul style="list-style-type: none"> <li>✓ main untouched</li> <li>✓ Documented: README, the Guide's step 6, and the reference</li> </ul>                                                                                                                                                                                                                            |
| ✓ | <b>H14 Blackbox names &amp; aliases</b>                                                                                                                                                                                                                                | Unit; v2 §5                                                                                                                                                                                                                                                                                                                                                        |
| ✓ | <ul style="list-style-type: none"> <li><code>fox_blackbox</code>, <code>/blackbox</code> routes, MCP and SDK names</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>✓ Every original name still works</li> </ul>                                                                                                                                                                                                                                                                                |
| ✓ | <b>H15 Policy gate (warn / block)</b>                                                                                                                                                                                                                                  | Unit; v2 §6, §6b                                                                                                                                                                                                                                                                                                                                                   |
| ✓ | <ul style="list-style-type: none"> <li>BBX02 warn / BBX01 block with JSON detail</li> <li>Blocked attempts recorded</li> <li>Per-rule <code>db_admin</code> override</li> </ul>                                                                                        | <ul style="list-style-type: none"> <li>✓ CLI, REST, MCP, web page; custom rules</li> <li>✓ Patterns match the running statement</li> <li>✓ TRUNCATE (its own guardrail, H4)</li> </ul>                                                                                                                                                                             |
| ✓ | <b>H16 Agent provenance &amp; execute_change</b>                                                                                                                                                                                                                       | Unit; v2 §7                                                                                                                                                                                                                                                                                                                                                        |
| ✓ | <ul style="list-style-type: none"> <li>Session, task, parent session on every agent change (also through the Gateway DSN)</li> <li>MCP <code>execute_change</code> with policy preview</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>✓ <code>fox_blackbox_sessions</code>, REST</li> <li>! Provenance is what the agent says (not verified)</li> </ul>                                                                                                                                                                                                           |
| 🔗 | <b>H17 Impact analysis</b>                                                                                                                                                                                                                                             | Unit; v2 §8                                                                                                                                                                                                                                                                                                                                                        |
| ✓ | <ul style="list-style-type: none"> <li>Dependents (views, foreign keys, indexes, depth)</li> <li>Other branches that have the object</li> <li>Policy verdict and risk score</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>✓ CLI, REST, MCP, inside <code>execute_change</code></li> <li>🔗 Functions, types, schemas not recognised from SQL</li> </ul>                                                                                                                                                                                                |
| ✓ | <b>H18 Blackbox diff, and promoting a branch's changes</b>                                                                                                                                                                                                             | Unit; v2 §8, §8b                                                                                                                                                                                                                                                                                                                                                   |
| ✓ | <ul style="list-style-type: none"> <li>Fork point, changes only on each side</li> <li>Objects changed on both</li> <li>Diff: CLI, REST, MCP</li> <li>Promotion: a branch's recorded statements, reviewed, then applied to another branch in one transaction</li> </ul> | <ul style="list-style-type: none"> <li>✓ Refused on a conflict (both sides changed the same object), on a statement the target's gate blocks, and when there is nothing to promote</li> <li>✓ Approving needs the right to manage the target, so an agent cannot approve its own request</li> <li>✓ Promotion: CLI, REST, MCP and the web Requests page</li> </ul> |

## I Import & migration

13 of 13 done

| ID   | Feature · sub-features                                                                                                                          | Tested by                                                                                                                      |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| ✓ I1 | <b>From Postgres</b>                                                                                                                            | Manual; no automated test yet                                                                                                  |
| ✓    | <ul style="list-style-type: none"> <li>Schema and data via <code>pg_dump</code></li> </ul>                                                      | <ul style="list-style-type: none"> <li>✓ Zero-table result reported as an error</li> </ul>                                     |
| ✓ I2 | <b>From MySQL ≤5.7 / MariaDB</b>                                                                                                                | Manual; no automated test yet                                                                                                  |
| ✓    | <ul style="list-style-type: none"> <li><code>pgloader</code> with type mapping (multi-arch image)</li> </ul>                                    |                                                                                                                                |
| ✓ I3 | <b>From MySQL 8.x</b>                                                                                                                           | Manual; no automated test yet                                                                                                  |
| ✓    | <ul style="list-style-type: none"> <li>Native <code>mysqldump</code> path (<code>caching_sha2_auth</code>)</li> </ul>                           | <ul style="list-style-type: none"> <li>✓ Types, NULLs, unicode, PK</li> <li>→ Other indexes, foreign keys, defaults</li> </ul> |
| ✓ I4 | <b>From MongoDB</b>                                                                                                                             | VM §9, §10                                                                                                                     |
| ✓    | <ul style="list-style-type: none"> <li>Collection → typed table, exact name</li> </ul>                                                          | <ul style="list-style-type: none"> <li>✓ Nested objects / arrays kept as jsonb</li> </ul>                                      |
| ✓ I5 | <b>From a .sql dump</b>                                                                                                                         | Unit; VM §11                                                                                                                   |
| ✓    | <ul style="list-style-type: none"> <li>Runs the dump into a new branch</li> <li>Fails listing every failed statement (instance kept)</li> </ul> | <ul style="list-style-type: none"> <li>✓ Same for Postgres imports (I1)</li> <li>✓ Automated test</li> </ul>                   |

|   |                                                                                                                                                                                                                                                                                                 |                      |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| ✓ | <b>I6 From JSON / NDJSON</b><br>✓ Columns inferred from top-level keys<br>✓ An invalid document fails the import, naming its element or line                                                                                                                                                    | Unit; VM §10, §11    |
| ✓ | <b>I7 From CSV</b><br>✓ Table from the header, exact column names<br>! Typed columns (all text today)                                                                                                                                                                                           | Manual               |
| ✓ | <b>I8 Continuous replication &amp; cutover</b><br>✓ Initial copy + streaming (logical replication)<br>✓ Cutover keeps data (CLI)<br>✓ Status and cutover from REST and the Import page<br>✓ A replicating branch is never auto-suspended<br>✓ Automated test (VM §5b) and driven in the browser | Unit; VM §5b         |
| ✓ | <b>I9 Guard handling during import</b><br>✓ Guardrail paused, recording stays on<br>✓ Re-armed even after a crash                                                                                                                                                                               | Manual               |
| ✓ | <b>I10 Live progress (CLI &amp; SSE)</b><br>✓ Live log and progress bar<br>! psql subprocess output not streamed                                                                                                                                                                                | Manual               |
| ✓ | <b>I11 Import REST endpoints</b><br>✓ From a connection string or an uploaded file<br>✓ Target names validated (the import creates the branch through branch.Create)<br>✓ Private, local and container sources for admins only; metadata refused                                                | Manual; unit; VM §1d |
| ✓ | <b>I12 Web Import page</b><br>✓ Database and file panels with live console<br>✓ No longer claims a SQLite loader that doesn't exist                                                                                                                                                             | Manual               |
| ✓ | <b>I13 Exact identifier fidelity</b><br>✓ Case kept ( <code>userId</code> , <code>leaseAiChats</code> )<br>! Field order of JSON sources not kept                                                                                                                                               | Unit; VM §10         |

## J ETL pipelines (experimental)

4 of 4 done

| ID   | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                            | Tested by   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| ✓ J1 | <b>Pipeline engine</b><br>✓ Extract → raw schema → SQL models → tests, in a fresh branch<br>✓ <code>ref()</code> / <code>source()</code> templating<br>→ <a href="#">Dependency graph</a><br>→ <a href="#">Incremental models</a><br>→ <a href="#">Scheduling</a><br>✓ A run replaces only a branch the pipeline made<br>✓ API sources are connection strings, never server paths | Unit; VM §9 |
| ✓ J2 | <b>Data-quality tests</b><br>✓ <code>not_null</code> , <code>unique</code> , <code>accepted_values</code> , <code>row_count_min</code> , <code>custom</code><br>✓ Failure → exit 1, data kept                                                                                                                                                                                     | Unit; VM §9 |
| ✓ J3 | <b>Stored pipelines &amp; run history</b><br>✓ Saved per user, run over the API, history<br>→ <a href="#">Read a run's log</a><br>→ <a href="#">Mark runs whose process died</a>                                                                                                                                                                                                  | Manual      |
| ✓ J4 | <b>Pipeline editor (web)</b><br>✓ Create, edit, run, results, history<br>→ <a href="#">Confirm before delete</a>                                                                                                                                                                                                                                                                  | Manual      |

## K APIs, SDKs & agents

6 of 6 done

| ID   | Feature · sub-features                                                                                                                                                                                                                                                                         | Tested by                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| ✓ K1 | <b>Control-plane REST API</b><br>✓ Branches, query, Blackbox, policy, impact, import, pipelines, keys<br>→ <a href="#">Versioned paths ( /api/v1 )</a><br>✓ Exact-origin CORS, branch-name check, body limits, timeouts<br>✓ Branches owned by their maker; others 404 (owner, admin, or main) | Unit; VM §1; v2 §0–§8; VM §1d |
| ✓ K2 | <b>OpenAPI spec</b><br>✓ Served, Blackbox 2.0 endpoints documented<br>✓ Matches the code (import fields, status codes, auth routes, /blackbox aliases)<br>✓ Drift check in CI                                                                                                                  | Unit (drift check)            |

|   |                                                                                                                                                                                                   |                                                                                                                                                                  |                            |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| ✓ | <b>K3 Python SDK</b><br>✓ Branches, query, Blackbox<br>→ <a href="#">Import, pipelines, keys, agents</a><br>✓ Contract tests against a live engine (nightly sdks suite)                           | → <a href="#">Publish to PyPI</a><br>✓ Contract tests against a live engine                                                                                      | Manual; VM sdks            |
| ✓ | <b>K4 TypeScript SDK</b><br>✓ Same client for Node and browsers<br>✓ Contract tests against a live engine (nightly sdks suite)                                                                    | → <a href="#">Publish to npm</a><br>✓ Compiles, and contract tests against a live engine                                                                         | Manual; VM sdks            |
| ✓ | <b>K5 Agent Branch API</b><br>✓ A branch per agent, own non-superuser role<br>✓ Provenance (task, parent, session)<br>✓ DSN reachable from the host (gateway-routed, branch-scoped key)           | ✓ Cap, listing and TTL count suspended branches<br>✓ Superuser switch only in <code>-tags insecure</code> builds<br>✓ Agents and their keys belong to the caller | Unit; VM \$5; v2 \$1c, \$7 |
| ✓ | <b>K6 MCP server</b><br>✓ Branches, run_sql, Blackbox tools<br>✓ policy_check, execute_change, impact, diff<br>✓ Requires an API key and acts as that account; a scoped key is held to its branch | ✓ Client config & 16-tool reference (docs/mcp.md)<br>✓ Same ownership rule as the REST API                                                                       | Unit; v2 \$1c–\$8          |

## L Web console

11 of 11 done

| ID   | Feature · sub-features                                                                                                                                                                                                                                                                          | Tested by                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ L1 | <b>Embedded single-origin UI</b><br>✓ Served by the engine at :8080                                                                                                                                                                                                                             | Manual                                                                                                                                                                                                                                                              |
| ✓ L2 | <b>Login &amp; signup</b><br>✓ Log in / sign up tabs, OAuth buttons                                                                                                                                                                                                                             | Manual; VM ui                                                                                                                                                                                                                                                       |
| ✓ L3 | <b>Dashboard</b><br>✓ Live tiles<br>✓ Branch table: create, suspend, resume, delete, copy DSN                                                                                                                                                                                                   | Manual; VM ui                                                                                                                                                                                                                                                       |
| ✓ L4 | <b>SQL console &amp; browser</b><br>✓ Schema sidebar, rows, structure, indexes<br>✓ Values shown as Postgres writes them (uuid, interval, inet, bytea)<br>✓ SQL editor<br>✓ System tables hidden by default, with a show button                                                                 | ✓ Automated UI tests (Playwright, in the nightly ui suite)<br>✓ Covered by the console's Playwright tests<br>✓ Covered by the console's Playwright tests<br>Manual; API in v2 \$6                                                                                   |
| ✓ L5 | <b>Blackbox page</b><br>✓ Entries per branch with badges<br>✓ Table / risk / date filters, and "Clear filters"<br>✓ Verify button; a row expands to its statement and session<br>✓ Live: new entries arrive on their own (3s), marked briefly, pausable; never while older pages are being read | ✓ The chain is visible: entry id, its row hash and the previous entry's, each copyable<br>✓ "Branch from before this change" on the row itself ( <code>main</code> ; explained elsewhere)<br>✓ Covered by the console's Playwright tests<br>v2 \$5b; browser; VM ui |
| ✓ L6 | <b>API keys page</b><br>✓ Create, copy, list, revoke<br>✓ A new key is shown inside its connection string (E3's next step)                                                                                                                                                                      | ✓ Covered by the console's Playwright tests<br>Manual; browser review 21 Sep; VM ui                                                                                                                                                                                 |
| ✓ L7 | <b>Landing, Guide &amp; Docs pages</b><br>✓ Landing, developer guide, reference                                                                                                                                                                                                                 | ✓ Refreshed against the code (https, sslmode, MCP, Blackbox, import routes)<br>Type-check + build                                                                                                                                                                   |
| ✓ L8 | <b>Blackbox integrity page</b><br>✓ Verify, checkpoint, export, branch-before<br>✓ Used signed in; every result matched the CLI                                                                                                                                                                 | ✓ New entries arrive on their own (3s), marked briefly, pausable<br>Type-check + build; browser review 16 Sep                                                                                                                                                       |

|   |                                              |                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |                                                                                       |
|---|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ✓ | <b>L9 Policies page</b>                      | <ul style="list-style-type: none"> <li>✓ Rules, warn/block, custom rules, preview, evaluations</li> <li>✓ Used signed in; matched the CLI, wrote only to the chosen branch</li> </ul>                                                         | <ul style="list-style-type: none"> <li>✓ Non-admin change refused with a readable message</li> <li>✓ Who can override: list, grant, revoke</li> </ul>                                                                                                                         | Type-check + build; API in v2 §6; browser review 16 Sep                               |
| ✓ | <b>L10 App sidebar layout</b>                | <ul style="list-style-type: none"> <li>✓ Every link opens its page, highlighted, with breadcrumb</li> </ul>                                                                                                                                   | <ul style="list-style-type: none"> <li>✓ Collapse persists across a reload; theme toggle persists</li> <li>✓ No console errors on any page</li> </ul>                                                                                                                         | Type-check + build; browser review 16 Sep; the under-860px drawer needs a manual look |
| ✓ | <b>L11 Design system &amp; FoxByte theme</b> | <ul style="list-style-type: none"> <li>✓ Tokens for colour, type, space, radius, elevation and motion; components use no raw values</li> <li>✓ Ember palette in both themes; filled controls legible ( <code>--accent-ink</code> )</li> </ul> | <ul style="list-style-type: none"> <li>✓ Fox mark from theme variables, wordmark from brand.json, favicon</li> <li>✓ Console, Blackbox, Policies and API keys re-laid out; one table style</li> <li>✓ Every page reviewed in both themes; console checked at 375px</li> </ul> | Type-check + build; browser review 21 Sep; not yet reviewed against real data         |

## M Build, CI & releases 7 of 7 done

| ID                                      | Feature · sub-features                                                                                                                                                                                                                                                                                                                                                                                                                               | Tested by                                                                                                             |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| ✓ <b>M1 Makefile targets</b>            | <ul style="list-style-type: none"> <li>✓ build, vet, test, integration suites</li> <li>✓ Suites run only in a throwaway test VM (guarded)</li> </ul>                                                                                                                                                                                                                                                                                                 | CI                                                                                                                    |
| ✓ <b>M2 Linux CI</b>                    | <ul style="list-style-type: none"> <li>✓ Build, vet, test, gofmt</li> <li>✓ Release script tests, checklist sync check</li> <li>✓ VM integration suites nightly in CI (one runner per suite)</li> </ul>                                                                                                                                                                                                                                              | CI                                                                                                                    |
| ✓ <b>M3 Windows CI</b>                  | <ul style="list-style-type: none"> <li>✓ Build, vet, test, Pester, installer encoding</li> </ul>                                                                                                                                                                                                                                                                                                                                                     | CI; no end-to-end WSL install test yet                                                                                |
| ✓ <b>M4 Releases &amp; GHCR image</b>   | <ul style="list-style-type: none"> <li>✓ Binaries for 5 platforms + fox-verify</li> <li>✓ SHA256SUMS merged safely between jobs</li> <li>✓ Fails unless <code>fox update</code>'s files are published</li> </ul>                                                                                                                                                                                                                                     | Script tests in CI                                                                                                    |
| ✓ <b>M5 WSL distro &amp; ZFS builds</b> | <ul style="list-style-type: none"> <li>✓ Prebuilt distro image</li> <li>✓ Downloaded and verified by the installer (seen on Windows)</li> <li>✓ Preloaded image tagged as the engine runs it</li> </ul>                                                                                                                                                                                                                                              | Actions artifacts; Pester; Windows run 16 Sep; the zstd unpack is unit-tested only, not run on a real Windows install |
| ✓ <b>M6 Postgres image</b>              | <ul style="list-style-type: none"> <li>✓ PostgreSQL 18 + wal-g for a fresh install; one image per supported major (16, 18)</li> <li>✓ The image follows the data: an install on 16 keeps running 16 after an update</li> <li>✓ A branch on the wrong major is refused before <code>docker run</code>, with the way out</li> <li>✓ The major lives in one place, and a test holds the Dockerfile, release matrix and Windows preload to it</li> </ul> | Unit; all three suites on 18                                                                                          |
| ✓ <b>M7 Product name in one file</b>    | <ul style="list-style-type: none"> <li>✓ brand.json → Go, shell, web and installer blocks, regenerated by <code>make brand</code></li> <li>✓ Stored names brand-free and frozen: a rename never migrates an install</li> <li>✓ Retired env prefixes, state directory, VM names and installs still handled</li> </ul>                                                                                                                                 | Unit (brand, gen, oldnames), in CI via go test                                                                        |

## N Not started (roadmap)

8 items

Planned or discussed, nothing built yet. When work starts, move the row into its area with the next id there.

| ID                                    | Item · scope                                                                                                           | Related                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> -            | <b>Multi-tenant projects</b><br><input type="checkbox"/> Projects / organisations, and roles within them               | G1, H6<br>✓ Per-branch permissions: owner or admin, on every branch-scoped route (G2 of the audit) |
| <input type="checkbox"/> -            | <b>Automatic HA failover</b><br><input type="checkbox"/> Health detection, fencing <input type="checkbox"/> Multi-host | D4                                                                                                 |
| <input type="checkbox"/> -            | <b>Pipelines: scheduling, incremental / CDC extract, DAG view</b>                                                      | J1                                                                                                 |
| <input type="checkbox"/> -            | <b>More import sources: Cassandra, DynamoDB</b>                                                                        | I1–I8                                                                                              |
| <input type="checkbox"/> -            | <b>pgvector in the Postgres image</b>                                                                                  | M6                                                                                                 |
| <input checked="" type="checkbox"/> - | <b>Release signatures</b> — done 22 Sep 2026                                                                           | A9, M4                                                                                             |
| <input type="checkbox"/> -            | <b>Hosted cloud tier</b>                                                                                               | —                                                                                                  |
| <input type="checkbox"/> -            | <b>Networked storage disaggregation</b> (README roadmap; see the storage engine note on page 1)                        | C1, D1                                                                                             |

### Out of scope by decision

A custom storage engine (PostgreSQL's own engine is used, for native transaction speed), sharding, Raft / distributed transactions, multi-region, HTAP.

FOX\_Checklist · keep in step with FOX\_Feature\_Implemented — CI checks the ids match